

Mth100

Final Term File

Quiz subjective and definitions

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By Iqra Azhar

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Q1. Write Pythagorean identity (2) marks

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

.Q 2 Find the value of cos(75) by trigonometric table. 5 marks

$$\cos(45+30)=\cos[a]\cos[b]-\sin[a]\sin[b]$$

$$\cos 75=\cos 45 \cos 30-\sin 45 \sin 30$$

$$\cos 75=1/\sqrt{2} [\sqrt{3}/2]-1/\sqrt{2}[1/2]$$

$$\cos 75=1/\sqrt{2}[\sqrt{3}-1]/2$$

$$\cos 75=[\sqrt{3}-1]/2\sqrt{2}$$

Q3. Find the standard deviation 36,45,14,18,20,24,28,33,47,55,51.. 5 marks

Numbers: 36, 45, 14, 18, 20, 24, 28, 33, 47, 55, 51

Step 1: Mean (μ) = (36 + 45 + 14 + 18 + 20 + 24 + 28 + 33 + 47 + 55 + 51) / 11 $\approx$ 32.09

Step 2: Squared differences:

$$(36 - 32.09)^2 \approx 15.14 \quad (45 - 32.09)^2 \approx 166.24 \quad (14 - 32.09)^2 \approx 328.79 \quad (18 - 32.09)^2 \approx 198.05$$

$$(20 - 32.09)^2 \approx 146.74 \quad (24 - 32.09)^2 \approx 65.64 \quad (28 - 32.09)^2 \approx 16.82 \quad (33 - 32.09)^2 \approx 0.83$$

$$(47 - 32.09)^2 \approx 221.42 \quad (55 - 32.09)^2 \approx 525.94 \quad (51 - 32.09)^2 \approx 357.28$$

Step 3: Mean of squared differences = (15.14 + 166.24 + 328.79 + 198.05 + 146.74 + 65.64 + 16.82 + 0.83 + 221.42 + 525.94 + 357.28) / 11 $\approx$ 193.62

Step 4: Standard Deviation = $\sqrt{193.62} \approx 13.91$

Q5 Find the slop of the (4,6)(2,1)

$$m=y_2-y_1/x_2-x_1$$

$$m=1-6/2-4$$

$$m=-5/-2$$

$$m=5/2$$

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Q6 $\tan x / \sin x = \sec x$

$$[\sin x / \cos x] * [1 / \sin x]$$

$$1 / \cos x$$

Sec x proved

Q7 $\sin 2A = 2 \sin A \cos A$

$$\sin(2A) = 2 * \sin(A) * \cos(A)$$

To prove this identity, we'll use the sum-to-product trigonometric identities:

$$\sin(A + B) = \sin(A) * \cos(B) + \cos(A) * \sin(B)$$

$$\sin(2A) = \sin(A + A)$$

$$\sin(A + A) = \sin(A) * \cos(A) + \cos(A) * \sin(A)$$

$$\sin(A + A) = \sin(A) * \cos(A) [1 + 1]$$

$$\sin(A + A) = 2 * \sin(A) * \cos(A)$$

Q.8 if multiple of 4 is set of A and B is set of 2 multiples then set A is subset of B or B is subset of A???

A is the set of all multiples of 4: $A = \{4, 8, 12, 16, \dots\}$

B is the set of all multiples of 2: $B = \{2, 4, 6, 8, 10, 12, \dots\}$

Now, we need to check whether A is a subset of B or B is a subset of A.

A set A is a subset of another set B if every element of A is also an element of B.

To determine whether A is a subset of B, we need to check if every element of A is also in B.

Looking at the sets, we can see that every element of A is also an element of B. Therefore, A is a subset of B.

To check whether B is a subset of A, we need to see if every element of B is also in A.

$$A = \{4, 8, 12, 16, \dots\} \quad B = \{2, 4, 6, 8, 10, 12, \dots\}$$

In this case, B contains elements that are not in A (e.g., 2, 6, 10, ...). Therefore, B is not a subset of A.

In conclusion A is a subset of B.

B is not a subset of A.

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Q9 Frequency density find out if width is 5 and frequency is 23

Frequency Density = Frequency / Width

Frequency Density = 23 / 5

Frequency Density $\approx$ 4.6

So, the frequency density in this case is approximately 4.6.

Q10 mode is 30 then and median 38 then find the mean..?

$3\text{median} = \text{mode} + 2\text{mean}$

$3[38] = 30 + 2\text{mean}$

$114 = 30 + 2\text{mean}$

$114 - 30 = 2\text{mean}$

$86 = 2\text{mean}$

Mean = $86/2$

Mean = 43

Q11 if frequency=12 n frequency density=0.6 then find the class width..?

Class Width = Frequency / Frequency Density

Frequency = 12 Frequency Density = 0.6

Class Width = $12 / 0.6$

Class Width = 20 So, the class width in this case is 20.

Q12 find the slope of the line through the (3,-1) and (-4,2)

Slope = $m = \frac{y_2 - y_1}{x_2 - x_1}$

$m = \frac{2 - (-1)}{-4 - 3}$

$m = \frac{3}{-7}$

Q13 the 3rd and the 4th terms of a geometric progression are 12 and 8 respectively. Find the sum to infinity of the progression.

. Let's assume that the first term is 'a' and the common ratio is 'r'.

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The 3rd term of the GP is given as 12, which means:

$$a * r^2 = 12$$

The 4th term of the GP is given as 8, which means:

$$a * r^3 = 8$$

Now, we can divide the second equation by the first equation to find the common ratio 'r':

$$(a * r^3) / (a * r^2) = 8 / 12$$

$$r = 8 / 12$$

$$r = 2/3$$

Now that we have the common ratio 'r', we can use it to find the first term 'a' using the first equation:

$$a * (2/3)^2 = 12$$

$$a * 4/9 = 12$$

$$a = 12 * 9 / 4$$

$$a = 27$$

So, the first term 'a' is 27, and the common ratio 'r' is 2/3.

Now, to find the sum to infinity (S) of the geometric progression, we can use the formula:

$$S = a / (1 - r)$$

$$S = 27 / (1 - 2/3)$$

Now, let's calculate the sum:

$$S = 27 / (1/3)$$

$$S = 27 * 3$$

$$S = 81$$
 So, the sum to infinity of the geometric progression is 81.

Question: 14 In how many ways can a team of 6 candidates can be formed out of a total of 10 candidates such that two particular candidates should be included in each department?

Step 1: Select the two particular candidates who must be included in each team. There are 2 ways to do this.

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Step 2: Now, we need to select the remaining 4 candidates from the remaining $10 - 2 = 8$ candidates. Since order doesn't matter, we will use combinations. The number of ways to choose 4 candidates from 8 is denoted as "8 choose 4" and can be calculated using the combination formula:

$C(n, r) = n! / (r! * (n - r)!)$ where n is the total number of candidates and r is the number of candidates we want to select. $C(8, 4) = 8! / (4! * (8 - 4)!)$

$$C(8, 4) = 70$$

Step 3: Now, multiply the results from Step 1 and Step 2 to get the total number of ways:

$$\text{Total number of ways} = \text{Step 1} * \text{Step 2}$$

$$\text{Total number of ways} = 2 * 70$$

$$\text{Total number of ways} = 140$$

So, there are 140 ways to form a team of 6 candidates, including two particular candidates in each team, out of a total of 10 candidates.

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1) **Define a set?** Definition: A set is an unordered collection of distinct objects. Objects in the collection are called elements of the set.

2) **What is roster method of sets?** The roster method of specifying a set consists of surrounding the collection of elements with braces. For example the set of counting numbers from 1 to 5 would be written as $\{1, 2, 3, 4, 5\}$.

3) **Define null set?** Definition: The set with no elements is called the empty set or the null set.

4) **Define universal set?** Definition: The universal set is the set of all things pertinent to a given discussion and is designated by the symbol U

5) **Define subset?** Definition: The set A is a subset of the set B , denoted $A \subset B$, if every element of A is an element of B .

6) **Define equal set?** Definition: Two sets A and B are equal if $A \subset B$ and $B \subset A$. If two sets A and B are equal we write $A = B$ to designate that relationship.

7) **Define intersection of sets?** Definition: The intersection of two sets A and B is the set containing those elements which are elements of A and elements of B . We write $A \cap B$ to denote A Intersection B . Example: If $A = \{3, 4, 6, 8\}$ and $B = \{1, 2, 3, 5, 6\}$ then $A \cap B = \{3, 6\}$

8) **Define union of sets?** Definition: The union of two sets A and B is the set containing those elements which are elements of A or elements of B . We write $A \cup B$ to denote A Union B . Example: If $A = \{3, 4, 6\}$ and $B = \{1, 2, 3, 5, 6\}$ then $A \cup B = \{1, 2, 3, 4, 5, 6\}$.

9) **What are algebraic properties of sets?** Or difference between commutative associative and distributive law? Commutative: Union and intersection are commutative operations. In other words, $A \cup B = B \cup A$ and $A \cap B = B \cap A$ Associative: Union and intersection are associative operations. In other words, $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = B \cap (A \cap C)$ Distributive: Union and Intersection are distributive with respect to each other. In other words $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

10) **Define cardinality?** With two types? Definition: Cardinality refers to the number of elements in a set A finite set has a countable number of elements An infinite set has at least as many elements as the set of natural numbers

11) **Define complex number?** Definition: Numbers of the form $a + bi$ are called complex numbers. a is the real part and b is the imaginary part. The set of complex numbers is denoted by C

12) **Define absolute value?** Definition: The absolute value or modulus of a complex number is the distance the complex number is from the origin on the complex plane.

13) **Define relation?** Definition: A mapping between two sets A and B is simply a rule for relating elements of one set to the other. A mapping is also called a relation.

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14) **Define domain and range?** Definition: The set consisting of members of the pre-image or inputs of a function is called its domain. For a given domain the set of possible outcomes or images of a function is called its range.

15) **Define even and odd function?** Definition: A function is called an even function if its graph is symmetric with respect to the vertical axis, and it is called an odd function if its graph is symmetric with respect to the origin.

16) **Define quadratic function?** Definition: A function of the type $y = ax^2 + bx + c$ where a , b , and c are called the coefficients, is called a quadratic function.

17) **Define a matrix?** Definition: A matrix is a rectangular arrangement of numbers in rows and columns. The order of a matrix is the number of the rows and columns. The entries are the numbers in the matrix. 18) **Define identity matrix?** Definition: A Square matrix with ones on the diagonal and zeros elsewhere is called an identity matrix. It is denoted by I .

19) **When a matrix has echelon form?** A matrix is in echelon form if it has the following properties Every non-zero row begins with a 1 (called a leading 1) Every leading one in a lower row is further to the right of the leading one above it. If there are zero rows, they are at the end of the matrix

20) **When a matrix has reduced echelon form?** A matrix is in reduced echelon form if in addition to the above three properties it also has the following property: Every other entry in a column containing a leading one is zero
Methods for finding Solutions of Equations: Using Row Operations: Recall that when we are solving simultaneous equations, the system of equations remains unchanged if we perform the following operations: Multiply an equation by a non-zero constants Add a multiple of one equation to another equation Interchange two equations.

21) **Define adjoint or adjugate?** Definition: Given a matrix A , calculate all the cofactors of A . We then form the matrix (of the cofactors. The Adjoint or Adjugate of A is the transpose of the matrix of the cofactors.

22) **Define sequence and terms?** Definition: Rows of numbers are called sequences, and the separate numbers are called terms of the sequence.

23) **Define arithmetic sequences?** Definition: An Arithmetic Sequence (or Arithmetic Progression) is a sequence in which each term after the first term is found by adding a constant, called the common difference (d), to the previous term

24) **Define geometric sequences?** Definition: A sequence in which each term after the first is found by multiplying the previous term by a constant value called the common ratio, is called a Geometric Sequence (or Geometric Progression). The formula for finding any term of a geometric sequence is $a_n = ar^{n-1}$

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25) Define geometrics series? Definition: A Geometric Series is the sum of the terms in a arithmetic sequence. The formula for finding the sum of the first n terms of a geometric sequence is given by $S_n = a(1 - r^n) / (1 - r)$

26) Define convergent and divergent? Definition: If a sequence of numbers approaches (or converges) to a finite number, we say that the sequence is convergent. If a sequence does not converge to a finite number it is called divergent.

27) What is multiplication principal? Multiplication Principle: If two operations A and B are performed in order, with n possible outcomes for A and m possible outcomes for B, then there are $n \times m$ possible combined outcomes of the first operation followed by the second

28) Describe Pascal's Triangle? Expression= Coefficients $(x + y)^1 = x + y = 1 \ 1$ $(x + y)^2 = x^2 + 2xy + y^2 = 1 \ 2 \ 1$ $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 = 1 \ 3 \ 3 \ 1$ $(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4 = 1 \ 4 \ 6 \ 4 \ 1$ $(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5 = 1 \ 5 \ 10 \ 10 \ 5 \ 1$ **29) What are Properties of Gradient?** The bigger the gradient's magnitude is, the steeper the line segment. Negative gradient means line is facing downwards. Positive gradient means the line is facing upwards. The slope gives the average rate of change in y per unit change in x , where the value of y depends on x . Two line segments that are parallel will have the same slope.

30) What are Properties of the Cosine Function? The cosine function has "period" 360° as it repeats itself after each revolution of 360° . $\cos(-\theta) = \cos \theta$ as the x -coordinate of P doesn't change when we reflect across the x -axis. $\cos(180^\circ - \theta) = -\cos \theta$ as the x -coordinate changes signs when reflected across the y -axis. $\cos(-180^\circ) = -\cos \theta$ as the x -coordinate changes signs when reflected across the origin. $\cos \theta$ is positive in the first and the fourth quadrant (as the x coordinate of P is positive), and negative in the second and the third quadrant as the x -axis is negative there. The range of the cosine function is between -1 and 1 . The maximum value of 1 is taken when $\theta = 0^\circ, \pm 360^\circ, \pm 720^\circ, \dots$, and the minimum value of -1 is at $\theta = \pm 180^\circ, \pm 540^\circ, \dots$

31) Define periodic function and period? The functions with the property that they keep repeating themselves are called periodic functions. The smallest interval for which the function repeats itself is called its period

32) What are Properties of the Sine Function? $\sin(-\theta) = -\sin \theta$ as the y -coordinate of P changes sign when we reflect across the x -axis. $\sin \theta$ is positive in the first and the second quadrant (as the y coordinate of P is positive), and negative in the third and the fourth quadrant as the y coordinate is negative there. $\sin(180^\circ - \theta) = \sin \theta$ because as we reflect across the y -axis the y -coordinate doesn't change. $\sin(-180^\circ) = -\sin \theta$ as the y -coordinate changes signs when reflected across the origin. Like the cosine function, the sine function is also periodic, with period 360° degrees, and range between -1 and 1 .

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33) What are Properties of the Tangent Function? $\tan(-) = -\tan$ as the y-coordinate of P changes sign when we reflect across the x-axis but the x-coordinate doesn't change sign. $\tan$ is positive in the first and the third quadrant (as the x and y coordinates of P have the same signs in these quadrants), and negative in the second and the fourth quadrant as the x and y coordinates have opposite signs in these quadrants. $\tan(180 -) = -\tan$ as the x-coordinate of P changes sign when we reflect across the y-axis but the y-coordinate doesn't change sign. The domain of $\tan$ does not include the angles for which x is 0, namely, for $\theta = \pm 90, \pm 270, \dots$. Like the cosine and sine functions, the tangent function is also periodic, but its period is 180. i.e. $\tan(\theta + 180) = \tan \theta$ and $\tan(\theta - 180) = \tan \theta$.

34) Define amplitude? Definition: The Amplitude of a function is the height from the mean (or the rest) value of the function to its maximum or minimum value.

35) Define inverse sin function? Definition: The inverse sine function is defined by $y = \arcsin x$ if and only if $\sin y = x$. The domain of $y = \arcsin x$ is $[-1, 1]$. The range of $y = \arcsin x$ is $[-\pi/2, \pi/2]$.

36) Define inverse cosin function? Definition: The inverse cosine function is defined by $y = \arccos x$ if and only if $\cos y = x$. The domain of $y = \arccos x$ is $[-1, 1]$. The range of $y = \arccos x$ is $[0, \pi]$.

37) Define inverse tangent function? Definition: The inverse tangent function is defined by $y = \arctan x$ if and only if $\tan y = x$. The domain of $y = \arctan x$ is $\mathbb{R}$. The range of $y = \arctan x$ is $(-\pi/2, \pi/2)$.

38) Define identity and conditional equation? Definition: Two functions f and g are said to be identically equal if $f(x) = g(x)$ for every value of x for which both functions are defined. Such an equation is referred to as an identity. An equation that is not an identity is called a conditional equation.

39) Define statistics? Definition: Methods of collection, organization and analysis of numerical information are collectively called statistics.

40) Define variable and data? Pieces of numerical and non-numerical information are called data. In order to collect data, you need to observe or measure some property; this property is called a variable.

41) Define quantitative and qualitative variables? Definition: A variable is qualitative if it is not possible for it to take a numerical value. A variable is quantitative if it can take a numerical value. A quantitative variable which can take any value in a given range is called a continuous variable. A quantitative variable which has clear steps between its possible values is called a discrete variable.

42) Define Descriptive Statistics? comprises those methods concerned with collection and describing a set of data so as to yield meaningful information is called descriptive statistics.

43) Define inferential statistics? Inferential Statistics comprises those methods concerned with analysis of a subset of data leading to predictions or inferences about the entire set of data.

44) Define population? Definitions: A population is defined as the set of all possible members of a stated group. A crosssection of the returns of all of the stocks traded on the New York Stock Exchange (NYSE) is an example of a population.

45) Define sample? Definition: A sample is defined as a subset of the population of interest. Once a population has been defined, a sample can be drawn from the population, and the sample's characteristics can be used to describe the population as a whole.

46) Define perimeter? Definition: A measure used to describe a

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characteristic of a population is referred to as a parameter. 47) **What is Nominal scale?**

Observations are classified or counted with no particular order. It consists of assigning items to groups or categories. No quantitative information is conveyed and no ordering of the items is implied. Nominal scales are therefore qualitative rather than quantitative. Religious preference, race, and gender are all examples of nominal scales. 48) **What is Ordinal scale?** All observations are placed into separate categories and the categories are placed in order with respect to some characteristic. Differences between values makes no sense. Political parties on left to right spectrum given labels 0, 1, 2; restaurant ratings, etc, are examples of ordinal scales. 49) **Define Interval scale?** This scale provides ranking and assurance that differences between scale values are equal. Difference makes sense, but ratio doesn't; and there is no natural zero. temperature (C,F) and dates are examples of interval scale

50) **What is Ratio scale?** These represent the strongest level of measurement. In addition to providing ranking and equal differences between scale values, ratio scales have a true zero point as the origin. Height, weight, age and length are all examples of ratio scale.

51) **What is frequency distribution?** Definition: A frequency distribution is a tabular presentation of statistical data that aids the analysis of large data sets. Frequency distributions summarize statistical data by assigning it to specified groups, or intervals.

52) **Define relative distribution?** Definition: Relative frequency is calculated by dividing the frequency of each interval by the total number of observations. Simply, relative frequency is the percentage of total observations falling within each interval.

53) **Define cumulative frequency?** Definition: Cumulative Frequency is calculated by summing the frequencies starting at the lowest interval and progressing through the highest. Cumulative frequency for any given interval is the sum of the frequencies up to and including the given interval.

54) **What is bar chart?** Definition: A Bar chart graphically represents the data sets by representing the frequencies as heights of bars.

55) **What is histogram?** Definition: A Bar chart which represents continuous data is called a histogram if The bars have no spaces between them (though there may be bars with zero height, which could look like spaces). The area of each bar is proportional to the frequency. If all the bars have the same width, then the height is proportional to the frequency

56) **Define frequency density?** Definition: Frequency Density is defined as the ratio between the frequency of a class and the class width. i.e. $\text{Frequency Density} = \frac{\text{Frequency}}{\text{Class Width}}$

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57) Define arithmetic mean? Definition: The Arithmetic Mean is the sum of the observation values divided by the number of observations. It is the most widely used measure of central tendency, and is the only measure where the sum of the deviations of each value from the mean is always zero. The formula for calculating the arithmetic mean of n values is: $\frac{1}{n} \sum_{i=1}^n X_i$

58) Define probability space? Definition: A Probability Space or Sample Space is the set of all the possible outcomes in an experiment

59) What are Characteristics of a Probability Distribution? The probability of any event A lies between 0 and 1, i.e. $0 \leq P(A) \leq 1$. The sum of probabilities of all the mutually exclusive and exhaustive events in a probability distribution equals 1.

60) What is the probability of selecting a truck at random that has either air bags or bucket seats? **Solution:** The addition rule for probabilities is used to determine the probability of at least one event among two or more events occurring. The probability of each event is added and the joint probability (if the events are not mutually exclusive) is subtracted to arrive at the solution. $P(\text{air bags or bucket seats}) = P(\text{air bags}) + P(\text{bucket seats}) - P(\text{air bags and bucket seats})$
 $= (125/220) + (110/220) - (75/220) = 0.57 + 0.50 - 0.34 = 0.73$ or 73 percent. Alternative: $1 - P(\text{no airbag and no bucket seats}) = 1 - (60/220) = 73\%$

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